STEMI: Fibrinolysis Pathway

This guideline is for adults with ST elevation myocardial infarction (STEMI) where transport to a hospital with facilities for immediate PCI cannot clearly occur within 90 minutes of STEMI diagnosis.

- Transmit the 12 lead ECG to the destination hospital.
- Administer 300 mg of aspirin PO.
- Complete the **fibrinolytic therapy checklist** (below). Seek urgent clinical advice prior to fibrinolysis if any cautions are present.
- Ensure hypertension (a systolic BP greater than 180 mmHg or a diastolic BP greater than 110 mmHg) is controlled before fibrinolysis. See **controlling hypertension** (below) if hypertension is present.
- Administer fibrinolysis (see **dosing tables** below).
- If cardiac arrest occurs prior to fibrinolysis:
  - Commence resuscitation using the **'cardiac arrest'** guideline.
  - Administer IV tenecteplase (see **dosing tables** below) if ROSC does not occur within five minutes of commencing resuscitation.
- Administer an opiate IV as required for moderate to severe pain.
- Administer 0.4 mg GTN SL with caution and withhold GTN SL if signs of poor perfusion are present.
- Transport the patient to a hospital with the facilities for immediate PCI and contact the appropriate clinician (see **hospital facilities** below).
- Enroute to hospital:
  - Measure vital signs every ten minutes and continue to ensure hypertension (if present) is controlled.
  - Monitor the patient closely for signs of complications, particularly bleeding.
  - Repeat a 12 lead ECG every 30 minutes but do not routinely transmit these.

Fibrinolytic therapy checklist

- Seek clinical advice if any of the following are present:
  - Suspected aortic dissection.
  - Major surgery, severe trauma, or severe brain injury within the last six weeks.
  - Intracranial surgery within the last six months.
  - Ischaemic stroke within the last six months.
  - Previous intracerebral haemorrhage.

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- Known cerebral aneurysm, arteriovenous malformation, or tumour.
- More than ten minutes of CPR.
- Non-compressible vascular puncture within the last 24 hours.
- Internal bleeding within the last six weeks.
- Lumbar puncture or epidural insertion within the last six weeks.
- TIA within the last three months.
- Known bleeding disorder.
- Taking an anticoagulant.
- A systolic BP greater than 180 mmHg or a diastolic BP greater than 110 mmHg despite treatment.
- Known to be pregnant or less than two weeks postpartum.
- The time of symptom onset was greater than 12 hours ago.
- Requires assistance with their activities of daily living.
- Severe comorbidities that significantly limit their life expectancy.

Controlling hypertension

- Ensure an opiate has been administered IV, and be generous with dosing.
- If the systolic BP is greater than 180 mmHg or the diastolic BP is greater than 110 mmHg, commence the following treatment to maintain a systolic BP of 140-160 mmHg and a diastolic BP below 110 mmHg:
  - Administer 0.4 - 0.8 mg of GTN SL every five minutes, as required. Move to the next step if hypertension is not controlled within approximately 10 minutes of commencing GTN.
  - Continue to administer GTN SL and administer labetalol IV:
    - Administer a 20 mg bolus IV every five minutes, as required.
    - If more than three doses are required, commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
    - Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
    - Administer additional boluses of 20 mg IV every five minutes as required.
  - If labetalol is contraindicated, or the target BP is not met within 30 minutes of commencing labetalol, seek clinical advice to discuss further options.

Fibrinolysis if aged less than 75 years

- Administer tenecteplase IV as per the dosing table below.
- Administer enoxaparin SC as per the dosing table below.

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- Administer 300 mg of clopidogrel PO.
- Administer 30 mg of enoxaparin IV:
  - Dilute 100 mg to a total of 10 ml using 0.9% sodium chloride.
  - Administer 30 mg (3 ml) IV.

	Tenecteplase IV	Enoxaparin SC
< 60 kg	30 mg (6 ml)	60 mg (0.6 ml)
60-69 kg	35 mg (7 ml)	70 mg (0.7 ml)
70-79 kg	40 mg (8 ml)	80 mg (0.8 ml)
80-89 kg	45 mg (9 ml)	90 mg (0.9 ml)
≥ 90 kg	50 mg (10 ml)	100 mg (1 ml)

Fibrinolysis if aged greater than or equal to 75 years

- Administer tenecteplase IV as per the dosing table below.
- Administer enoxaparin SC as per the dosing table below.
- Administer 75 mg of clopidogrel PO.
- Do not administer enoxaparin IV.

	Tenecteplase IV	Enoxaparin SC
< 60 kg	15 mg (3 ml)	45 mg (0.45 ml)
60-69 kg	17.5 mg (3.5 ml)	50 mg (0.5 ml)
70-79 kg	20 mg (4 ml)	60 mg (0.6 ml)
80-89 kg	22.5 mg (4.5 ml)	70 mg (0.7 ml)
≥ 90 kg	25 mg (5 ml)	75 mg (0.75 ml)

Hospitals with the facilities for immediate PCI

- Auckland City Hospital:

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- Available 24/7.
- Call the cardiology registrar on 021 906624.
- HHStJ personnel: Lifepak 15 transmission site code NR01.

- Waikato Hospital:
  - Available 24/7.
  - Call the cardiology registrar on 07 839 8676.
  - HHStJ personnel: Lifepak 15 transmission site code CR01 or NR07.
- Tauranga Hospital:
  - Available if the patient will arrive in ED between 0730hrs and 1530hrs Monday to Friday (excluding public holidays).
  - Phone the cardiac catheter lab coordinator on 027 7047712.
  - HHStJ personnel: Lifepak 15 transmission site code CR04.
- Wellington Hospital:
  - Available 24/7.
  - WFA personnel: call the Cardiology Nurse Coordinator on 021 504746.
  - All other personnel: call the on call doctor for HHStJ via the Clinical Desk.
  - HHStJ personnel: Lifepak 15 transmission site code CLINICAL DESK.
- Nelson Hospital:
  - Availability is variable.
  - Call the on call cardiologist on 0800 728 4929, extension 2, to confirm availability prior to transport.
  - HHStJ personnel: Lifepak 15 transmission site code SI02.
- Christchurch Hospital:
  - Available 24/7.
  - Call the cardiology registrar on 0800 728 4929, extension 1.
  - HHStJ personnel: Lifepak 15 transmission site code SI01.
- Dunedin Hospital:
  - Available 24/7.
  - Call the on call cardiologist on 0800 728 4929, extension 3.
  - HHStJ personnel: Lifepak 15 transmission site code SI03.

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Backup

- Backup from an ICP/CCP should be requested if the patient has:
  - Bradydysrhythmia, or
  - Tachydysrhythmia, or
  - Signs of shock.

Additional information

General principles

- Fibrinolytic therapy must be initiated as soon as possible because outcomes are predominantly determined by the time to reperfusion of the coronary artery. For example, fibrinolytic therapy has a higher priority than administering pain relief or commencing extrication.
- If the patient is being transported to a medical facility for fibrinolytic therapy personnel should initiate staging via the Air Desk as soon as is feasible.
- Fibrinolytic therapy may be administered via IO access if IV access cannot be obtained, but IO access must be secure.
- Be prepared to treat cardiac arrest but do not routinely attach defibrillation pads unless the patient is being transported by helicopter. Cardiac arrest will occur in approximately 5% of patients, the initial rhythm will usually be ventricular fibrillation, and most patients will achieve return of spontaneous circulation after 1-2 shocks.
- Following fibrinolytic therapy, the patient should be transported to a hospital with facilities for immediate PCI, whenever this is feasible and safe, because approximately 30% of patients receiving fibrinolytic therapy will not achieve reperfusion and will require rescue PCI.
- Some patients with severe comorbidities may not be eligible for rescue PCI and this will be determined when seeking clinical advice following the checklist. In this setting patients should be transported to the most appropriate hospital, in discussion with the advising clinician.
- Have a low threshold for administering antiemetics.

The fibrinolytic therapy checklist

- The conditions within the checklist place the patient at increased risk of bleeding, and/or alter the risk/benefit balance, associated with fibrinolytic therapy.

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- Aortic dissection may involve the coronary arteries and cause STEMI. Aortic dissection usually presents with a sudden onset of severe pain that is maximal at the time of onset and often described as sharp, tearing or stabbing and may radiate to the back.
- Major surgery is any form of surgery where if subsequent bleeding developed, it could be life-threatening. Examples include surgery within a body cavity, surgery involving the spine and any form of joint replacement.
- Severe trauma is trauma severe enough to cause an injury where if subsequent bleeding developed, it could be life-threatening. Examples include rib fractures, intra-abdominal injury, and pelvic fractures.
- Severe brain injury is an injury that was severe enough to result in an injury visible on a CT scan.
- Intracranial surgery includes any operation or procedure involving the brain.
- Ischaemic stroke within the last six months includes thrombotic and embolic strokes, even if the patient was successfully treated.
- Previous intracerebral haemorrhage includes all forms of previous bleeding within the brain.
- A cerebral aneurysm is a known weakness in the wall of a cerebral artery causing dilation of the vessel. If the patient has not heard the term 'cerebral or brain aneurysm' it is highly unlikely the patient has one.
- An arteriovenous (AV) malformation is an abnormal connection between an artery and a vein. The vein is exposed to arterial pressure because there is no capillary bed to reduce the pressure so the vein dilates and may rupture. If the patient has not heard of the term 'AV malformation' it is highly unlikely the patient has one.
- A cerebral tumour is any form of brain tumour, primary or metastatic.
- Non-compressible vascular puncture includes any invasive procedure involving a non-compressible site, for example an organ biopsy.
- Internal bleeding includes gastrointestinal bleeding, bleeding from the urinary tract and bleeding from the lung. 'Coffee ground' vomit is very non-specific and is not a reliable clinical sign of gastrointestinal bleeding. Normal menstrual bleeding is not considered to be internal bleeding.
- A lumbar puncture is a procedure in which a needle is inserted between two lumbar vertebrae, most commonly to collect cerebrospinal fluid for diagnostic testing. An epidural insertion is a procedure in which a needle is inserted between two vertebrae, most commonly for the placement of a local anaesthetic agent.
- Anticoagulants include medicines such as warfarin, dabigatran, rivaroxaban and apixaban, but not antiplatelet medicines such as aspirin, clopidogrel and ticagrelor.

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- Hypertension must be controlled before and after fibrinolysis because uncontrolled hypertension increases the risk of intracranial haemorrhage following fibrinolysis. Paramedics should seek clinical advice regarding labetalol administration if an ICP/CCP is not immediately available and GTN administration has not successfully decreased the patient's blood pressure.
- Severe comorbidities that significantly limit their life expectancy include dementia, severe COPD, severe heart failure, a requirement for aged residential care and metastatic cancer with weight loss.

Informed consent

- Written consent is not required for fibrinolytic therapy, but the patient needs to provide informed consent. For example, inform the patient that:
  - They are having a 'heart attack' and that this is life-threatening.
  - The 'heart attack' is due to a clot blocking one of the blood vessels in their heart.
  - A medicine will be administered to help break down the clot and this reduces the chance they will die or be left with permanent heart damage.
  - The earlier the medicine is administered, the more likely it is to work.
  - Rarely the medicine can cause bleeding, for example inside the body. Very rarely this bleeding can cause death but the risk of this is very low.
  - The medicine is more likely to help them than to harm them.

Complications following fibrinolytic therapy

- Treat the patient using the appropriate guideline if complications from STEMI, for example dysrhythmia, occur.
- Seek clinical advice if complications from fibrinolytic therapy occur, for example bleeding.

Response and/or transport by helicopter

- Time to a PCI-capable hospital is generally less critical following fibrinolysis, however rescue PCI may be required in some cases.
- Response by helicopter may be indicated if a CCP is required for the management of a dysrhythmia or cardiogenic shock, and helicopter arrival at scene is at least 15 minutes faster than a road response.
- Transport by helicopter may be indicated if it will save more than 30 minutes compared to road transport.
- For helicopter transport criteria to be met, the road transport time will usually need to be more than 90 minutes.

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More Information

Version history

3 August 2026 1.1.0.6
Update: Anticoagulant treatment for fibrinolysis.

16 June 2026 1.1.0.1
Update: Treatment guidance updated.

11 September 2023 1.0.4
New: Guideline redeveloped.

23 February 2023 1.0.2.1
Update: Fixed drug volume error in tenecteplase volume for >90kg patient.